

I. Project Introduction

1. Overview

1.1 Project Information

- **Project name:** Design and Development of an Inventory, Borrowing and Hall Booking Management System for FPT University Ho Chi Minh City
- **Project code:** SU26SE109
- **Group name:** GSU26SE41
- **Software type:** Web
- **Production URL:** <https://fi2m.site/>

1.2 Project Team

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2. Product Background

In a dynamic educational environment like FPT University Ho Chi Minh City, the rapid growth of the student population alongside an expanding slate of academic and extracurricular activities has placed immense operational strain on administrative departments. Currently, managing physical assets and campus event spaces relies heavily on manual interventions, decentralized email communications, or unoptimized software platforms. This fractured operational approach introduces several systemic bottlenecks:

- **Fragmented Asset Tracking:** Monitoring equipment such as televisions, projectors, tables, and chairs is often restricted to generic, high-level categories rather than serialized tracking numbers. This lack of granular visibility creates severe accountability gaps, making it difficult to pinpoint responsibility when equipment is misplaced or damaged. Furthermore, retrieving history logs from unstructured email records is highly time-consuming and prone to human error.
- **Resource Scheduling Conflicts:** The manual validation of reservation requests for university halls and common facility areas frequently leads to booking overlaps. These

double-booking conflicts disrupt student club activities, lecture arrangements, and formal academic ceremonies.

- **The Digital-to-Physical Gap:** Campus stakeholders (including lecturers, staff members, and student clubs) waste significant windows of time physically verifying asset status or looking for room availability since the university lacks a unified data repository to check digital asset tracking statuses.
- **Lack of Automated Oversight:** Monitoring compliance metrics, such as identifying late equipment returns, tracking structural asset damage, or coordinating preventive maintenance cycles, relies entirely on manual tracking. Without automated notifications, turnaround times for asset redeployment remain slow, and leadership lacks long-term cost predictability.

To resolve these compounding issues, the administrative staff and university community requested a unified solution. This project introduces a centralized, automated web management system designed to seamlessly manage inventory tracking, track check-out/check-in workflows, and provide live, conflict-free space booking models custom-tailored for FPT University HCM.

3. Existing Systems

To establish an optimized architecture for this application, existing commercial resource enterprise platforms and institutional management systems were analyzed to identify industry-standard features, design choices, and functional limitations.

3.1 ServiceNow

ServiceNow is a cloud-based enterprise platform that provides IT service management, asset management, workflow automation, and other business process management capabilities. Its IT Asset Management (ITAM) and Hardware Asset Management (HAM) solutions support organizations in managing hardware and other assets throughout their lifecycle.

System Actors: IT/Asset Administrators, Employees, and Service Desk Staff.

Key Features:

- **Asset Lifecycle Management:** Supports the management of assets throughout their lifecycle, including requests, fulfillment, deployment, maintenance, and retirement.
- **Hardware Asset Management:** Provides capabilities for tracking and managing hardware assets, including their location, status, ownership, and lifecycle information.
- **Inventory Management:** Supports stockroom and warehouse inventory management to maintain accurate asset records and availability.
- **Request and Fulfillment Workflows:** Provides workflows for requesting, provisioning, and fulfilling assets.
- **Workflow Automation:** Automates asset-related processes and tasks to reduce manual work and improve operational consistency.
- **Asset Visibility and Tracking:** Provides centralized visibility into asset information, including where assets are located, who is using them, and their current lifecycle status.
- **Mobile Capabilities:** Hardware Asset Management provides mobile capabilities for managing assets and asset-related tasks.
- **Pros:**
 - **Comprehensive asset lifecycle management:** Provides a centralized platform for managing assets from request and fulfillment through maintenance and retirement.

- **Strong workflow automation:** Configurable workflows can automate asset requests, fulfillment, inventory operations, and other administrative processes.
- **Centralized asset visibility:** Provides a unified view of hardware and other managed assets, helping organizations track asset status, location, and usage.
- **Enterprise scalability:** ServiceNow is designed as an enterprise platform and can integrate asset management with other business workflows and systems.
- **Cons:**
 - **Not specialized for equipment borrowing:** Although ServiceNow supports asset requests and fulfillment, its standard asset management model is broader than a dedicated equipment borrowing system. Specific workflows such as borrower approval, equipment handover, return processing, and borrowing extensions may require additional configuration or customization.
 - **Implementation complexity:** ServiceNow provides a broad enterprise platform with extensive configuration and workflow capabilities, which can introduce additional implementation and maintenance overhead for a smaller institutional system.
 - **Licensing requirements:** Some asset management capabilities, such as Hardware Asset Management, are provided as licensable applications or packages, which may make the platform less suitable for small-scale projects with limited budgets.
 - **Customization for institution-specific requirements:** University-specific requirements, such as custom borrowing rules, quantity-based versus serial-based inventory management, specialized handover procedures, or incident tracking for borrowers, may require additional configuration or custom application development.

Link URL: <https://www.servicenow.com/>

3.2 Skedda

Skedda is a cloud-based space booking and scheduling platform designed for managing shared workspaces, meeting rooms, desks, facilities, and other bookable resources. It provides features such as interactive floor plans, real-time availability, booking calendars, configurable booking rules, access controls, and integrations with external calendar and identity systems. Skedda also supports use cases in universities and academic laboratories.

- **System Actors:** Space Administrators, Facility Managers, and Space/Resource Lecturers.
- **Key Features:**
 - **Interactive Floor Plans:** Lecturers can view spaces through interactive maps and check real-time availability before making a booking.
 - **Booking Calendar:** Provides Day, Month, List, Grid, and Map views for managing and monitoring bookings.
 - **Booking Rules and Permissions:** Administrators can configure booking conditions, quotas, booking windows, buffer times, access restrictions, and approval requirements.
 - **Conflict Prevention:** Real-time availability and scheduling rules help prevent conflicting and double bookings.

- **Resource Booking:** Skedda can be used to schedule not only spaces but also other bookable resources and equipment, particularly in laboratory and academic environments.
- **Calendar and System Integrations:** Supports integrations with platforms such as Microsoft 365 and Google Workspace, as well as SSO and other workplace systems.
- **Utilization Analytics:** Provides insights into space usage, booking patterns, and utilization to support facility management decisions.
- **Pros:**
 - **Strong scheduling capabilities:** Interactive floor plans, real-time availability, and calendar views provide an intuitive way to manage shared spaces and resources.
 - **Flexible booking policies:** The rules and roles engine allows administrators to enforce booking windows, quotas, permissions, and approval policies automatically.
 - **Self-service booking:** Lecturers can independently find available spaces or resources and create bookings without requiring administrators to manually process every reservation.
 - **Suitable for academic environments:** Skedda explicitly supports universities, academic labs, classrooms, and other educational facilities.
- **Cons:**
 - **Not specialized for equipment lifecycle management:** Although Skedda can schedule resources and equipment, its primary focus is booking and space management rather than detailed equipment lifecycle management, physical handover, return processing, or asset custody tracking.
 - **Limited maintenance and incident management:** Equipment maintenance, physical damage records, repair workflows, and borrower incident histories are not the core focus of the platform and may require integration with another system.
 - **Institution-specific integration requirements:** While Skedda provides integrations and SSO capabilities, connecting bookings with an institution's internal asset management, inventory, or activity systems may still require additional integration work.
 - **Different operational focus:** Skedda is optimized for scheduling and utilization of spaces and resources, whereas a dedicated equipment borrowing system can provide deeper workflows for request approval, warehouse fulfillment, equipment replacement, handover confirmation, return processing, and borrower incident tracking.

Link URL: <https://www.skedda.com/>

4. Business Opportunity

The current administrative environment at FPT University Ho Chi Minh City presents a highly valuable digital transformation opportunity. By replacing outdated, manual processes with a single, tailored resource management ecosystem, the university can optimize internal resource allocation, save substantial administrative labor hours, and pioneer a modern campus management framework.

At present, many resource-related activities can involve multiple administrative steps, including submitting requests, checking resource availability, obtaining approvals, coordinating handovers,

tracking returns, and managing maintenance information. When these activities rely heavily on manual coordination or disconnected tools, staff may spend significant time verifying information, updating records, and communicating with different stakeholders. This creates opportunities for duplicated work, inconsistent data, delayed processing, and limited visibility into the current status of campus resources.

The proposed system provides an opportunity to centralize these activities into one integrated platform. Lecturers can submit and track resource requests digitally, while administrators can review requests, manage approvals, monitor resource availability, and coordinate the allocation process through a unified interface. By maintaining a centralized record of resource transactions, the system can also provide greater transparency throughout the resource lifecycle, from allocation and handover to return and maintenance.

Another important business opportunity lies in improving resource utilization. A centralized resource management system can provide administrators with clearer information about which resources are available, currently allocated, being returned, or undergoing maintenance. This visibility can support more informed allocation decisions and reduce unnecessary conflicts between resource requests. It can also help the university identify frequently requested resources and areas where resource capacity may need to be adjusted.

The system can further support preventive and corrective maintenance by maintaining structured records of equipment conditions, maintenance requests, repair activities, and resource status. Instead of treating maintenance as a separate administrative activity, integrating it into the resource lifecycle allows the university to obtain a more complete view of each asset and its operational history. This can contribute to better planning and more consistent resource management.

From an operational perspective, automation can reduce repetitive administrative tasks and allow staff to focus on higher-value activities. Automated validation, status tracking, notifications, and standardized workflows can make processes more consistent while reducing the need for repeated manual communication. For Lecturers, this can result in a clearer and more convenient experience when requesting or returning university resources.

The opportunity also extends to data-driven decision-making. As resource transactions are recorded digitally, the university can potentially analyze usage patterns, request frequency, resource availability, maintenance activities, and other operational indicators. These insights can support future decisions regarding resource procurement, allocation policies, maintenance planning, and operational improvements.

Furthermore, the proposed solution establishes a foundation for future digital transformation initiatives across the campus. Once resource management processes are standardized and digitized, the platform can potentially be extended with additional capabilities such as advanced reporting, intelligent recommendations, automated notifications, and integration with other university systems. This makes the project more than a solution for replacing manual procedures; it can serve as a foundation for developing a more connected and data-driven campus management environment.

Overall, the business opportunity is centered on transforming resource management from a fragmented and administrative-heavy process into a centralized, transparent, and scalable digital workflow. The expected organizational value includes improved operational efficiency, better resource visibility, more structured maintenance management, reduced repetitive work, and stronger support for data-driven decision-making. This aligns with the broader objective of digital transformation at FPT University Ho Chi Minh City and provides a practical foundation for modernizing internal campus operations.

5. Software Product Vision

The vision for the Management System for FPT University Ho Chi Minh City is to establish a unified, secure, and smart web-based digital ecosystem that transforms campus operations. The software will bridge the clear operational gap between digital management systems and real-world institutional physical resources. By consolidating multiple fragmented operational pillars into a single platform, the software will eliminate traditional manual overhead and establish complete institutional transparency.

6. Project Scope & Limitations

The system will strictly govern the digital logistics, scheduling workflows, and physical tracking mechanics of campus assets. The development scope is bounded by three major functional modules:

- **SKU-Level Inventory Management:** Developing an administration portal for warehouse managers to register, catalog, and update individual campus assets (TVs, tables, chairs, tech equipment) with unique tracking serials rather than generic categories.
 - Real-time lifecycle logging for physical items, mapping their status across statuses such as Available, Borrowed, In Maintenance, or Damaged.
- **Automated Borrowing & Return Workflow:**
 - Building a self-service request pipeline allowing verified lecturers, staff, and student club leads to view active catalog availability, submit reservation requests, and track approval states.
 - An automated system engine to track return deadlines, auto-flag overdue handovers, and document physical item conditions during the check-in process.
- **Intelligent Venue Booking Matrix:**
 - Creating a live scheduling calendar showcasing real-time occupancy configurations for university halls, lobbies, and common spaces.
 - Implementing server-side constraint checks that block overlapping reservation timelines to systematically eliminate double-bookings.

Project Limitations (Out-of-Scope)

To ensure structural viability and timely deployment within the capstone cycle, the following parameters are explicitly excluded from this system's development scope:

- **Limited Guest Access:** External public Lecturers cannot book university halls or borrow internal equipment. Only Lecturers authenticated through the university's recognized internal accounts can interact with reservation pipelines.

6.1 Major Features

[FE-01] Authentication & Lecturer Management

- **FE-01.1:** Sign in(Admin, Lecturer roles).
- **FE-01.2:** Forgot password, reset password, and account setup completion
- **FE-01.3:** Admin Lecturer management and bulk import of accounts via Excel
- **FE-01.4:** Personal profile (/me) and avatar management

[FE-02] Warehouse & Asset Management

- **FE-02.1:** Equipment catalog: create / update / soft-delete, with image uploads
- **FE-02.2:** Category and brand management
- **FE-02.3:** Inventory tracking in two modes:
 - BY_SERIAL — individual SKU / QR units.
 - BY_QUANTITY — quantity-only tracking.
- **FE-02.4:** SKU statuses: AVAILABLE, RESERVED, BORROWED, MAINTENANCE, DAMAGED, LOST
- **FE-02.5:** SKU change history (SKU logs)
- **FE-02.6:** Inventory statistics dashboard
- **FE-02.7:** Stock import, quantity updates, and item status updates

[FE-03] Warranty Management

- **FE-03.2:** Warranty tracking by batch and/or equipment item
- **FE-03.3:** Filters for active, expiring soon, expired, and unconfigured warranties
- **FE-03.4:** Summary views and priority lists for admin follow-up

[FE-04] Equipment Audit

- **FE-04.1:** Create and manage audit sessions
- **FE-04.2:** Audit by scope: ALL / BY_SERIAL / BY_QUANTITY
- **FE-04.3:** Record discrepancies and losses (e.g., AUDIT_LOST)
- **FE-04.4:** Track session progress and final audit results

[FE-05] Repair Tickets

- **FE-05.1:** Create and manage equipment repair tickets
- **FE-05.2:** Link SKU status transitions (MAINTENANCE / DAMAGED) to the repair workflow
- **FE-05.3:** Track repairs through completion, permanent damage, or loss

[FE-06] Equipment Borrowing

- **FE-06.1:** Lecturer borrow lifecycle: DRAFT → PENDING → APPROVED → BORROWED → RETURNING → RETURNED
- **FE-06.2:** Terminal / special statuses: REJECTED, CANCELLED, NOT_COLLECTED, OVERDUE
- **FE-06.3:** Availability checks by borrow/return date window
- **FE-06.4:** Admin approve / reject / hand over / confirm return
- **FE-06.5:** Return with condition reporting (intact / damaged / lost) and quantity breakdown

- **FE-06.6:** Borrow extensions with schedule conflict handling
- **FE-06.7:** Urgent borrow — admin-created requests that start directly as BORROWED
- **FE-06.8:** Borrow request history (full lifecycle audit trail)
- **FE-06.9:** Automated schedulers: DRAFT promotion, pickup expiry (NOT_COLLECTED), OVERDUE detection, email reminders, extension revalidation

[FE-07] Overdue Handling & Asset Handover

- **FE-07.1:** Admin overdue borrow monitoring
- **FE-07.2:** Overdue borrower A can request handover to the next borrower B
- **FE-07.3:** B can accept or reject; supports both BY_SERIAL and BY_QUANTITY
- **FE-07.4:** Shortage handling after A→B handover

[FE-08] Borrower Incidents & Credibility

- **FE-08.1:** Incident logging: NOT_COLLECTED, OVERDUE, DAMAGED, LOST
- **FE-08.2:** Credibility score / penalty token system
- **FE-08.3:** Admin penalty review and restoration; blacklist when credibility is low
- **FE-08.4:** Blacklist warning emails

[FE-09] Venue & Hall Booking

- **FE-09.1:** Admin CRUD for venues (rooms / halls)
- **FE-09.2:** Self-service booking by Lecturers based on schedule availability
- **FE-09.3:** Check-in / check-out with no-show scheduling
- **FE-09.4:** Booking extensions (extend end time) with admin approve/reject
- **FE-09.5:** Admin venue & booking overview dashboard

[FE-10] Notifications

- **FE-10.1:** Dynamic email notifications (Thymeleaf templates) for borrow, booking, blacklist, and related events
- **FE-10.2:** In-app notifications plus realtime SSE delivery
- **FE-10.3:** Notification centers for both Admin and Lecturer

[FE-11] AI Insights (Gemini)

- **FE-11.1:** Warehouse statistics analysis (/ai/analyze-stats)
- **FE-11.2:** Warranty risk insights (/ai/warranty-insights)
- **FE-11.3:** Inventory health insights (/ai/inventory-insights)

[FE-12] Media & Platform Services

- **FE-12.1:** Image/media upload via Cloudinary
- **FE-12.2:** Standardized ApiResponse wrapper, pagination, and OpenAPI/Swagger docs
- **FE-12.3:** Soft-delete on entities; schema changes via Flyway migrations

6.2 Limitations & Exclusions

Limitations (LI):

LI-01: The browser never accesses the database. All reads/writes go Controller → Service → Repository → PostgreSQL.

LI-02: Status moves forward only on the happy path. There is no undo after submit, approve, handover, or return confirm.

LI-03: One authenticated Lecturer per request. The requester cannot act as admin; admin cannot complete the Lecturer return submit step.

LI-04: A new request starts as DRAFT. Admin cannot see it until it becomes PENDING.

LI-05: Edit / cancel is allowed only while DRAFT and inside the draft window (default 30 minutes from createdAt).

LI-06: Minimum borrow duration is 30 minutes (`expectedReturnDate - expectedBorrowDate`).

LI-07: Creating a draft occupies a time slot (`overlappingDemand`) but does not lock physical stock / SKU. `BorrowInventoryHelper` only reads `usableCapacity` and `remainingInWindow`.

LI-08: If remaining qty in the window is less than requested qty → 409 Conflict. The request is not saved.

LI-09: Blacklisted Lecturers and Lecturers over the penalty-token block cannot create a request.

LI-10: DRAFT → PENDING happens when the Lecturer submits, or when `BorrowDraftPromotionScheduler` runs (every 5 minutes after the window ends). Promotion is not instant.

LI-11: After PENDING, the Lecturer cannot edit or cancel. Inventory is still not reserved until admin confirms a line.

LI-12: Admin confirms per line, not the whole request in one call. Status becomes APPROVED only when every line is decided and at least one line is confirmed.

LI-13: Reservation on confirm is a time-slot plus SKU assign (`reserveDetailInventoryOnConfirm`). It is not a warehouse checkout yet.

LI-14: If some lines are still undecided, status stays PENDING.

LI-15: Handover is allowed only from APPROVED with `collected = true`.

LI-16: Status becomes BORROWED only when all confirmed lines are handed over. Otherwise it stays APPROVED.

LI-17: Warehouse shortage is auto-cut (quantity reduced and a note written). Handover is not blocked.

LI-18: The Lecturer may submit a return only from BORROWED (or OVERDUE, which is outside these diagrams).

LI-19: Return is per selected line, with photo and condition (BY_QUANTITY breakdown or BY_SERIAL SKU conditions). Inventory is not released yet. Status becomes RETURNING.

LI-20: A line with a pending return batch cannot be submitted again until admin confirms that batch.

LI-21: Admin confirms return per line, and only while status is RETURNING.

- LI-22:** Inventory is released only when admin confirms (releaseReturnedDetailInventory).
- LI-23:** If any units remain outstanding, status goes back to BORROWED, not RETURNED. RETURNED only when all confirmed lines are fully returned.
- LI-24:** BorrowInventoryHelper is not a repository. Persistence uses injected repositories and JPA dirty-checking inside the caller's @Transactional method.
- LI-25:** Diagrams show the synchronous HTTP path only. Email and in-app notifications run as side effects and are not drawn as participants.

Exclusions (EX):

- EX-01:** Urgent borrow — admin creates a request directly as BORROWED and skips DRAFT / PENDING / APPROVED.
- EX-02:** Update DRAFT (PUT /{id}) — edit inside the 30-minute window is not drawn.
- EX-03:** Cancel DRAFT (DELETE /{id}) — DRAFT → CANCELLED is not drawn.
- EX-04:** Disapprove line / REJECTED — admin reject-all plus inventory release is not drawn.
- EX-05:** NOT_COLLECTED — handover collected = false, or auto-expire after the 24-hour pickup grace.
- EX-06:** OVERDUE — scheduler BORROWED → OVERDUE after expectedReturnDate.
- EX-07:** A→B overdue handover — transfer from overdue borrower A to next borrower B.
- EX-08:** Resolve shortage — manual quantity cut on an APPROVED line.
- EX-09:** Borrow extension — Lecturer extend-return-date (preview, conflict, wait-DRAFT).
- EX-10:** Schedule conflicts — admin conflict groups and resolve UI.
- EX-11:** Borrower incidents — DAMAGED / LOST / NOT_COLLECTED records.
- EX-12:** Penalty and blacklist admin — token deduct, credibility, blacklist CRUD. Only the create-time eligibility check is in DRAFT.
- EX-13:** Email, in-app, and scheduler jobs — draft promotion, pickup expiry, overdue poll, reminder mails are not sequence participants.
- EX-14:** Venue booking — room/venue booking is a separate module.
- EX-15:** Payment, deposit, and late fee — no money flow in borrow.
- EX-16:** Guest / anonymous borrow — JWT Lecturer is required.
- EX-17:** Bulk confirm, bulk handover, and bulk return — APIs are one detailId per call.
- EX-18:** Undo / rollback — no reverse from APPROVED, BORROWED, RETURNING, or RETURNED.
- EX-19:** Waitlist, recurring borrow, and multi-warehouse — not implemented.
- EX-20:** Offline / barcode / QR pickup desk — handover is admin HTTP only.
- EX-21:** Real-time push (WebSocket) — Lecturer DRAFT/APPROVED screens poll; no live channel in these diagrams.
- EX-22:** Cloudinary / file storage internals — return images are uploaded, but storage is not a diagram participant.
- EX-23:** Partial warehouse handover edge cases — mixed A→B plus warehouse lines kept as APPROVED until all are handed over are not expanded here.
- EX-24:** Same-minute race on DRAFT overlap — internal exception in build Draft Items; not modelled as a Lecturer flow.